

Exp No: 6

Arduino to 16x2 LCD Display using Proteus

OBJECTIVES

- To interface a 16×2 LCD display with Arduino UNO using 4-bit communication mode.
- To write and upload an Arduino program to display text on the LCD.
- To simulate and verify the Arduino–LCD interfacing circuit in Proteus.

PROGRAM

```
#include <LiquidCrystal.h>

// RS, EN, D4, D5, D6, D7

LiquidCrystal lcd(7, 8, 9, 10, 11, 12);

void setup()
{
    lcd.begin(16, 2);    // Initialize 16x2 LCD
    lcd.print("Hello World!"); // Display message
}

void loop()
{
    lcd.scrollDisplayLeft(); // Scroll text to the left
    delay(700);             // Wait for 700 ms
}
```

OUTPUT

